

The Branch-and-Bound algorithm to solve Integer Programs

1 Introduction

In this lecture, we present the Branch-and-Bound algorithm, which lies at the core of all solvers for integer programming. Recall that the *linear relaxation* (LR) or *LP relaxation* of an integer program

$$\max c^T x : Ax \leq b, x \in \mathbb{Z}^n$$

is the LP obtained by dropping the integrality constraints:

$$\max c^T x : Ax \leq b.$$

2 Branch-and-Bound: explanation through examples

Observe that the optimal value of an integer programming problem in maximization form is less or equal to optimal value of its linear relaxation, since the feasible region of the latter contains the feasible region of the former. That is:

$$\max\{c^T x : Ax \leq b, x \in \mathbb{Z}^n\} \leq \max\{c^T x : Ax \leq b\}.$$

Moreover, the LR is a linear program, which is easy to solve.

Let now x^* be the optimal solution of the LR. If x^* is an integer vector, from what we saw above, it is optimal. Otherwise, it has at least a fractional component, say w.l.o.g. $x_1 = \delta \notin \mathbb{Z}$. Since our final optimal solution must be an integer vector, we know that, for all candidate solutions, one of the following must hold:

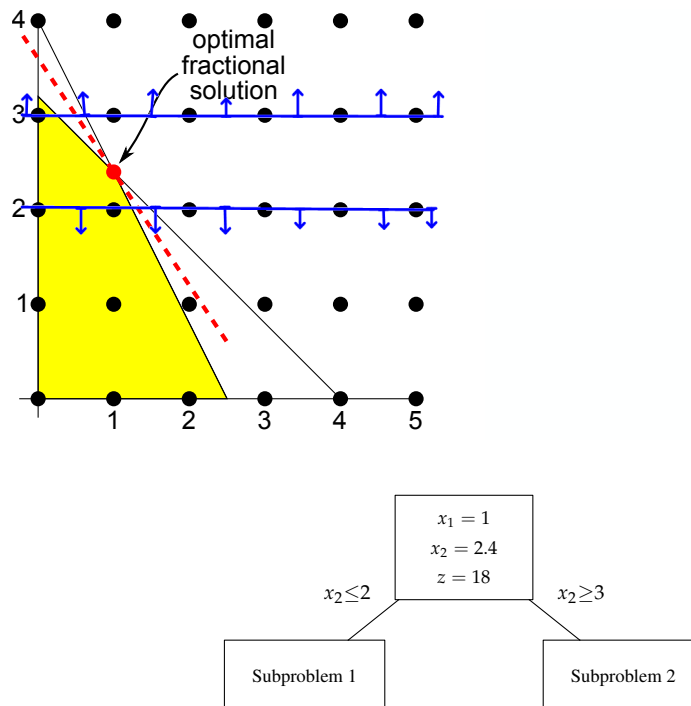
- (i) $x_1 \leq \lfloor \delta \rfloor$, or
- (ii) $x_1 \geq \lceil \delta \rceil$.

Hence, we split the original problem into two subproblems, to which exactly one of constraints (i) and (ii) has been added, and focus on the solution of the new subproblems instead of the original one: the best of the optimal solutions of those will be the optimal solution of the original problem.

Branch and bound can therefore be seen as a “smart enumeration process” that iteratively applies the observations above – and other, as we will see in an example – until it finds the optimal solution to the problem. We now illustrate this on an example:

$$\begin{aligned} \text{Max } & 6x_1 + 5x_2 \\ & 4x_1 + 5x_2 \leq 16 \\ & 8x_1 + 5x_2 \leq 20 \\ & 0 \leq x_1 \leq 2 \\ & 0 \leq x_2 \leq 3 \\ & x_1, x_2 \text{ integer} \end{aligned}$$

We solve the LP relaxation. Its optimal solution of the LP relaxation turns out to be $x_1 = 1, x_2 = 2.4$ (the LP relaxation can be solved by graphical method, or simplex method, or using a solver). The optimal solution is fractional, because $x_2 = 2.4$ is not an integer. We branch into $x_2 \leq 2$ and $x_2 \geq 3$.



Branching creates two subproblems, described below.

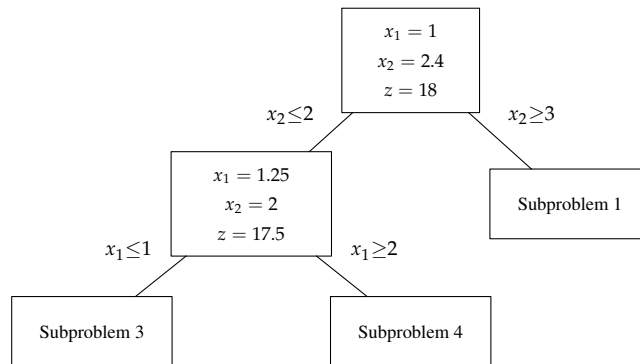
Subproblem 1:

$$\begin{aligned}
 &\text{Max } 6x_1 + 5x_2 \\
 &4x_1 + 5x_2 \leq 16 \\
 &8x_1 + 5x_2 \leq 20 \\
 &0 \leq x_1 \leq 2 \\
 &0 \leq x_2 \leq 2 \\
 &x_1, x_2 \text{ integer}
 \end{aligned}$$

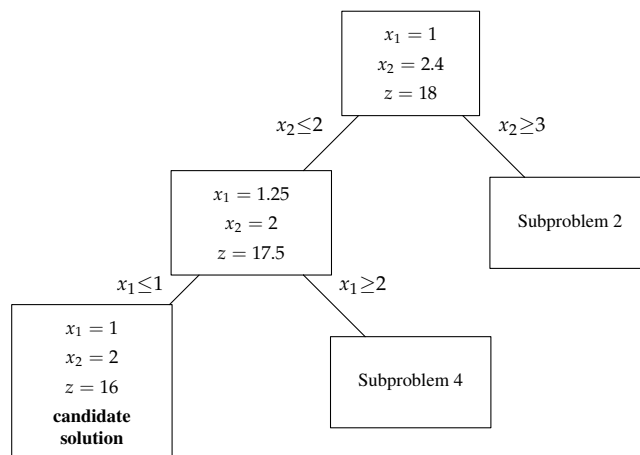
Subproblem 2:

$$\begin{aligned}
 &\text{Max } 6x_1 + 5x_2 \\
 &4x_1 + 5x_2 \leq 16 \\
 &8x_1 + 5x_2 \leq 20 \\
 &0 \leq x_1 \leq 2 \\
 &3 \leq x_2 \leq 3 \\
 &x_1, x_2 \text{ integer}
 \end{aligned}$$

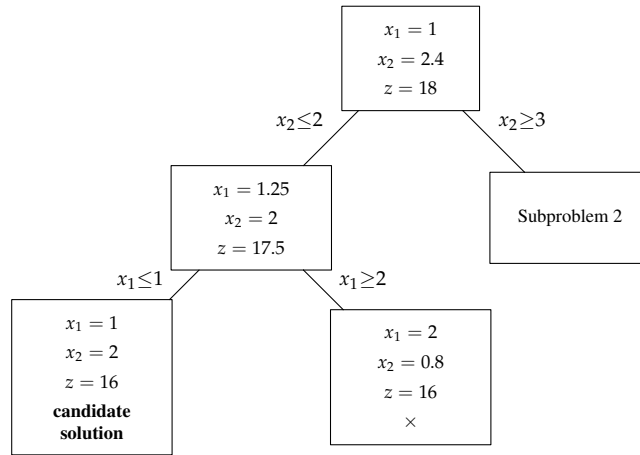
We choose to focus on **Subproblem 1**. The optimal solution of its linear relaxation is $x_1 = 1.25, x_2 = 2, z = 17.5$ is fractional, since $x_1 = 1.25$. We branch on $x_1 \leq 1$ and $x_1 \geq 2$. The updated Branching Tree is reported below.



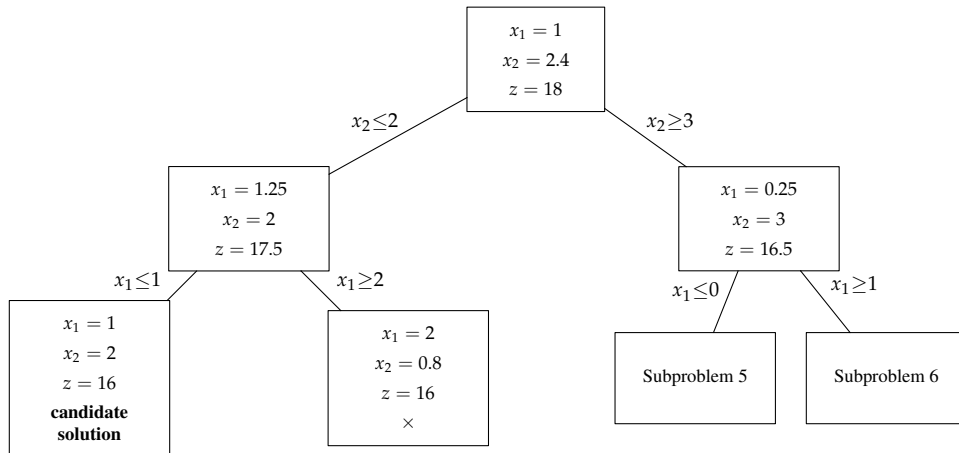
We choose to focus on **Subproblem 3**. The optimal solution of its LR turns out to be an **integer solution**: $x_1 = 1, x_2 = 2, z = 16$. Therefore, this is a candidate solution to the original integer problem. We **prune the node** (i.e., we do not create more subproblems from this subproblem). However, it is possible that better solutions can be obtained by solving other subproblems, so we need to investigate all subproblems that have been created but not solved yet. The updated branching tree is reported below.



Now we go back and choose **Subproblem 4**. The optimal solution of its LR is $x_1 = 2, x_2 = 0.8$. This is fractional, but its value is $z = 16$. Thus we **prune the node and do not branch**, since we already have a **candidate integer solution** of value $z = 16$. The updated branching tree is reported below.



Now we go back to **Subproblem 2**. The optimal solution of its LR turns out to be $x_1 = 0.25, x_2 = 3, z = 16.5$. This is fractional, but its value $z = 16.5$ is more than $z = 16$ of the candidate solution. It is still possible that a better solution can be found by branching¹. Since $x_1 = 0.25$, we branch on $x_1 \leq 0$ and $x_1 \geq 1$. The updated branching tree is given below:

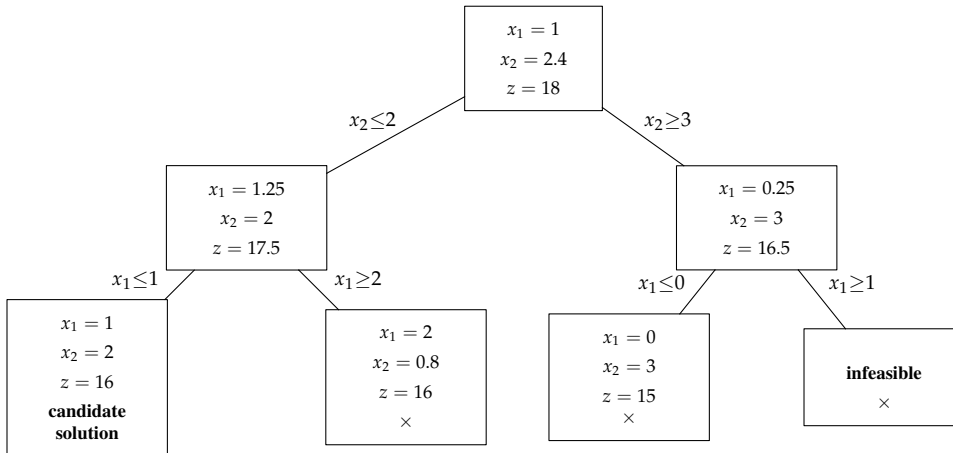


Let us now consider **Subproblem 5**. The optimal solution of its LR is $x_1 = 0, x_2 = 3$ with value $z = 15$ which is no better than our candidate solution. We prune the node.

We are left with **Subproblem 6**. When we solve its LR, we find out that it is infeasible. We prune the node.

We now have solved all subproblems. We can thus summarize the branching tree below.

¹Here we ignore the fact that optimal z should also be an integer, since it is the product of an integer cost vector an integer solution vector, otherwise we would not need to branch.



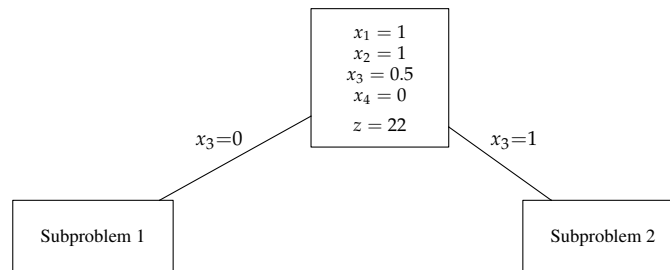
We conclude that the optimal integer solution is the best candidate solution found so far: $x_1 = 1$, $x_2 = 2$ of value $z = 16$. This concludes the algorithm.

Another example: Branch and bound for Knapsack

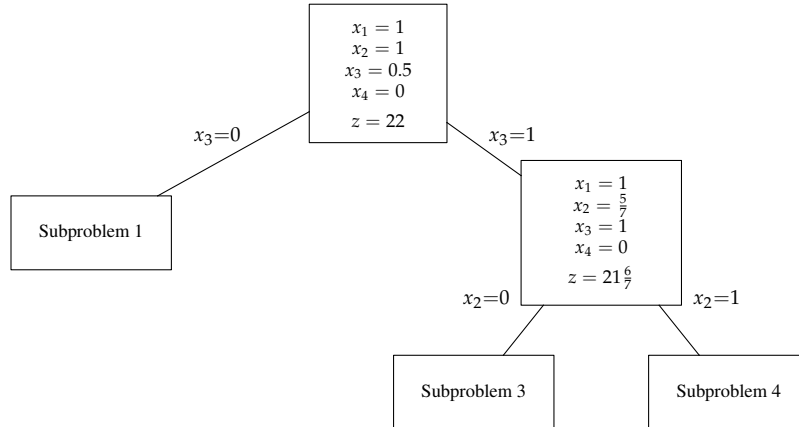
We see another example of Branch-and-bound on another integer problem, which is an instance of the classical **Knapsack** problem (we will study this problem in detail later on).

$$\begin{aligned} \text{Max } z &= 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ 5x_1 + 7x_2 + 4x_3 + 3x_4 &\leq 14 \\ x_1, x_2, x_3, x_4 &\in \{0, 1\} \end{aligned}$$

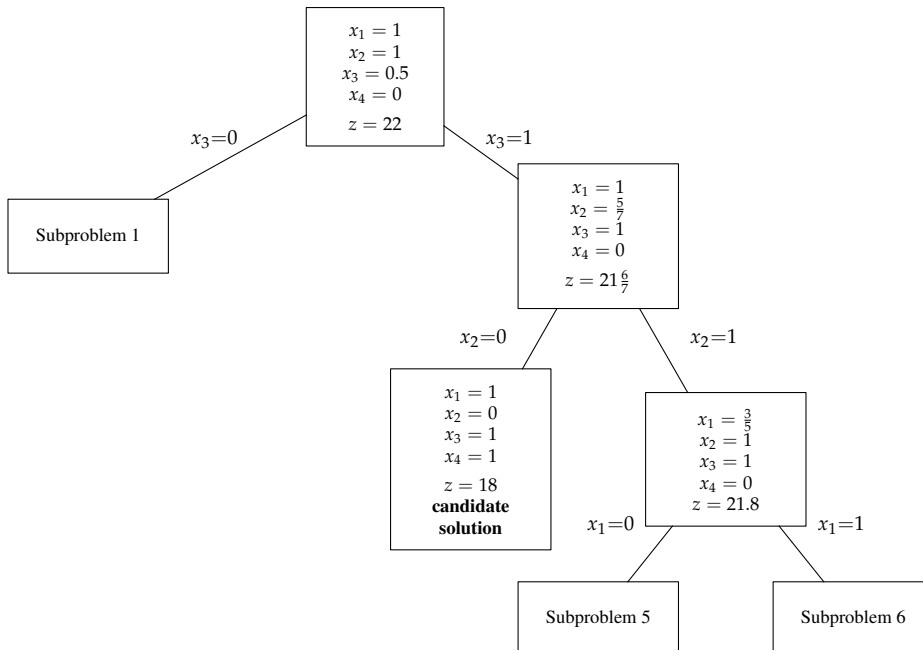
The optimal solution of the LR is given by $x_1 = 1, x_2 = 1, x_3 = 0.5, x_4 = 0$. It has objective function value 22. Since $x_3 = 0.5$, we branch on $x_3 \leq 0$ and $x_3 \geq 1$. Since we know that we must have $0 \leq x_3 \leq 1$ from the original IP, this branching is equivalent to setting $x_3 = 0$ and $x_3 = 1$, respectively.



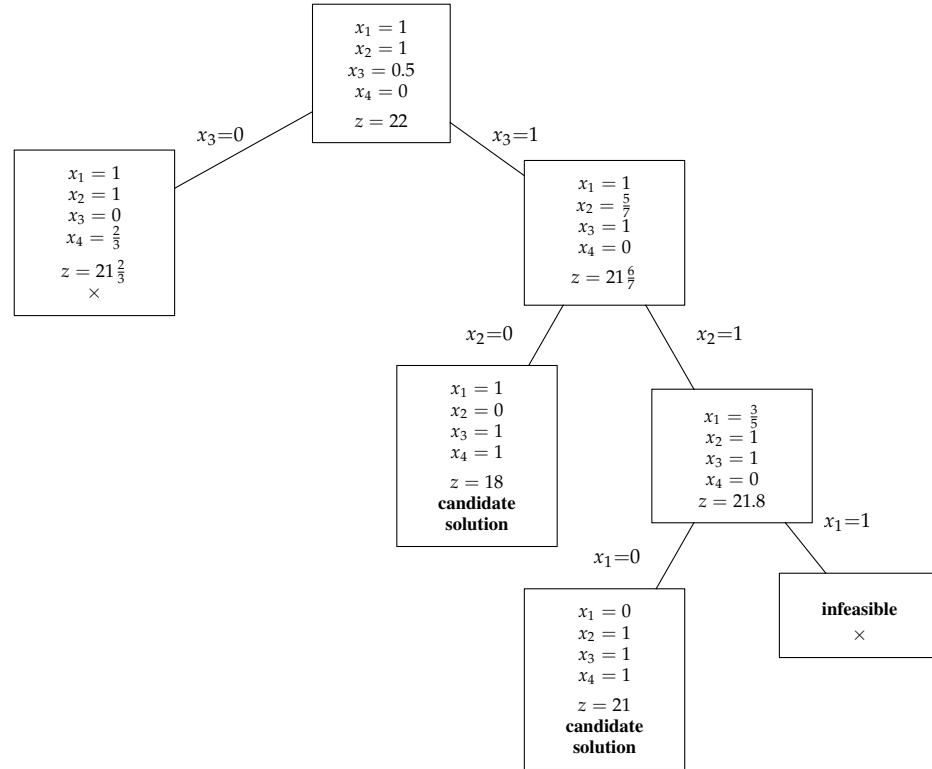
We decide to solve **Subproblem 2**. The optimal solution of its LR is $x_1 = 1, x_2 = 5/7, x_3 = 1, x_4 = 0$ of value $21 + \frac{6}{7}$. We branch on $x_2 = 0$ and $x_2 = 1$.



We choose to solve **Subproblem 3**; its LR has optimal solution $x_1 = 1, x_2 = 0, x_3 = 1, x_4 = 1$. It is an integer solution, thus a **candidate solution**, and it has value 18. We then solve **Subproblem 4**. Its LR has optimal solution $x_1 = 3/5, x_2 = 1, x_3 = 1, x_4 = 0$, and has value 21.8. We branch on $x_1 = 0$ and $x_1 = 1$. Here is an updated snapshot of the Branching tree.



We choose to solve **Subproblem 5**. The optimal solution of its LR is $x_1 = 0, x_2 = 1, x_3 = 1, x_4 = 1$. This is integer, so it is another candidate solution. Its objective function value is 21, so it is better than the solution found before. The LR of Subproblem 6 is infeasible. This concludes the exploration of the branching tree, which is summarized next.



The optimal solution is therefore the best candidate solution found, which is $x_1 = 0, x_2 = 1, x_3 = 1, x_4 = 1$. It has value 21.

3 Branch and bound: a summary

We can summarize the Branch-and-bound approach for an integer program in maximization form

$$\max c^T x : Ax \leq b, x \in \mathbb{Z}^n$$

as follows. At the beginning, we have:

- $v = -\infty, \bar{x} = \emptyset$ (no solution has been found so far);
- $\mathcal{S}$ contains exactly one problem, which is the original IP.

At every step, we keep the value v of the best candidate solution $\bar{x}$ found so far and a list $\mathcal{S}$ of subproblems to solve. While $\mathcal{S}$ contains at least one supproblem, we:

- Choose a supproblem P and remove it from $\mathcal{S}$.
- Solve its LR.
- If the LR is infeasible, **prune** the node of the branching tree. Else, let x^* be the optimal solution.

- If $c^T x^* \leq c^T \bar{x}$ is not larger than the profit of $\bar{x}$, **prune** the node of the branching tree.
- Else if $x^* \in \mathbb{Z}^n$: **prune** the node and set $\bar{x} = x^*$.
- Else, let $x_i^* = \alpha \notin \mathbb{Z}$. **Create two subproblems**, one by adding $x_i \leq \lfloor \alpha \rfloor$ to the current supproblem, and the other by adding $x_i \geq \lceil \alpha \rceil$ to the current subproblem.
- Add the new subproblems to $\mathcal{S}$.

We then output $\bar{x}$, the best solution found so far. This is an optimal solution to the original IP.